

Who was central on the web in April 2009?

A **10.9 GB** MemeTracker dump — 167M lines of posts and hyperlinks — collapsed to a **domain-level link graph** and scored with seven graph algorithms, all computed in Cypher / GDS. Nodes are registered domains; a weighted edge **A→B** counts every time a post on A linked to B.

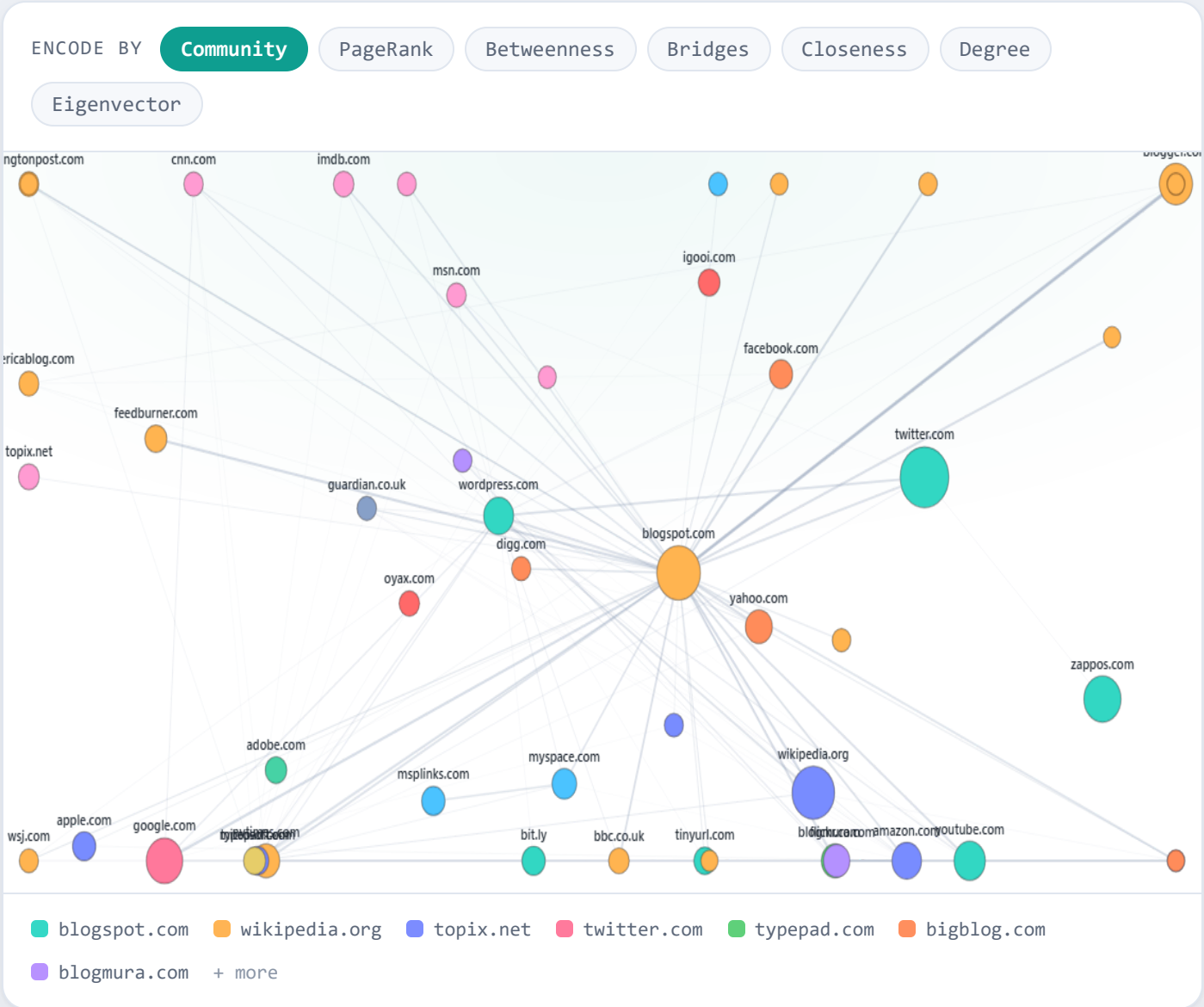
795k DOMAINS	2.21M WEIGHTED LINKS	23.55M RAW HYPERLINKS	10k CORE (DEG≥50)	3.1k COMMUNITIES
0.61 MODULARITY	524k BRIDGE EDGES			

How to read this — in one breath

Every dot is a **website** (a domain like `nytimes.com`), and every line means one site linked to another. We took **~795,000 sites** and ranked them **seven different ways** — each one a different answer to the same question: **who mattered most on the web in April 2009?** The panels below give each answer in plain English; the numbers underneath are the raw scores.

The map: who links to whom

top 50 domains by PageRank · size & colour switch by measure



Seven ways to be important

top 15 domains per measure, across all 795,268 nodes

Who connects separate worlds?

Bridges between otherwise-separate groups — traffic between communities passes through these sites.

Betweenness · sampled (2000 pivots), directed

01 🏆	blogspot.com		65.60M
02	wordpress.com		17.17M
03	google.com		6.07M
04	typepad.com		5.54M
05	flickr.com		4.96M
06	yahoo.com		2.59M
07	nytimes.com		2.57M
08	myspace.com		2.31M
09	youtube.com		2.08M
10	mahalo.com		2.00M
11	twitter.com		1.55M
12	adobe.com		1.48M
13	spiegel.de		1.34M
14	wsj.com		1.31M
15	live.com		1.30M

Who holds the network together?

Sit on the most “cut” links — remove them and whole chunks of the web break away.

Bridges · bridge edges incident to the domain

01 🏆	blogspot.com		126k
02	wordpress.com		24k
03	mahalo.com		9.1k
04	typepad.com		7.5k
05	ameblo.jp		4.9k
06	kinkaa.de		4.1k
07	articlesbase.com		3.5k
08	chacha.com		3.4k
09	craigslist.org		2.9k
10	emediawire.com		2.5k
11	web-log.nl		2.0k
12	splinder.com		1.9k
13	cocolog-nifty.com		1.7k
14	exblog.jp		1.6k
15	wikia.com		1.5k

● Who is closest to everyone?

Fewest hops, on average, to reach any other site.

Closeness · Wasserman-Faust, on the degree $\geq$ 50 core

01 🏆	wikipedia.org		0.654
02	blogspot.com		0.648
03	google.com		0.618
04	youtube.com		0.611
05	twitter.com		0.554
06	nytimes.com		0.547
07	wordpress.com		0.543
08	amazon.com		0.533
09	yahoo.com		0.530
10	flickr.com		0.507
11	bbc.co.uk		0.502
12	facebook.com		0.501
13	cnn.com		0.499
14	typepad.com		0.496
15	wsj.com		0.491

● Who links out the most?

Sheer outbound volume — the biggest firehoses of links.

Degree · weighted out-degree (Σ link weight)

01 🏆	blogspot.com		7.18M
02	wikia.com		690k
03	golivewire.com		636k
04	mashget.com		562k
05	articlesbase.com		493k
06	wordpress.com		440k
07	yunqueland.com		267k
08	ameblo.jp		236k
09	mahalo.com		231k
10	prolifeblogs.com		225k
11	feedsportal.com		212k
12	scouty.de		179k
13	rakuten.co.jp		173k
14	sportsnipe.com		166k
15	typepad.com		142k

Who is backed by important sites?

Linked by other influential sites, not just by many sites.

Eigenvector · directed, 46 iters (converged)

01	🏆 wikipedia.org		0.169
02	blogspot.com		0.168
03	google.com		0.147
04	youtube.com		0.144
05	twitter.com		0.124
06	nytimes.com		0.113
07	amazon.com		0.103
08	wordpress.com		0.098
09	yahoo.com		0.090
10	cnn.com		0.076
11	washingtonpost.co...		0.075
12	bbc.co.uk		0.075
13	wsj.com		0.074
14	facebook.com		0.072
15	msn.com		0.070

Who is most trusted overall?

Google-style PageRank — a link from an important place counts for more.

PageRank · weighted, damping 0.85

01	🏆 twitter.com		749.42
02	blogspot.com		561.24
03	wikipedia.org		526.22
04	zappos.com		362.24
05	google.com		344.87
06	blogger.com		261.51
07	youtube.com		221.45
08	wordpress.com		192.85
09	amazon.com		184.43
10	flickr.com		142.93
11	nytimes.com		141.03
12	yahoo.com		137.91
13	blogmura.com		128.84
14	myspace.com		100.69
15	msplinks.com		83.82

The web's neighbourhoods

Louvain communities

3,145 clusters • modularity 0.61 • 12 largest shown

The algorithm split the web into **3,145 clusters** that link within themselves far more than to the outside. Here are the 12 biggest, each labelled by its leading site.

	blogspot.com		196k • 24.7%
	wikipedia.org		125k • 15.7%
	topix.net		79k • 10.0%
	twitter.com		58k • 7.3%
	typepad.com		40k • 5.0%
	bigblog.com		31k • 3.8%
	blogmura.com		28k • 3.5%
	wikipedia.com		27k • 3.5%
	google.de		27k • 3.4%
	flickr.com		25k • 3.1%
	igooi.com		22k • 2.8%
	myspace.com		21k • 2.6%

All the numbers

every measure, as a number, for the top domains · click a column header to sort

DOMAIN	BETWEENNESS	BRIDGES	CLOSENESS	DEGREE	EIGENVECTOR	PAGERANK	COMMUNITY
twitter.com	1,553,344	571	0.5538	1,633	0.1241	749.42	#261016
blogspot.com	65,600,461	126,243	0.6478	7,176,350	0.1685	561.24	#32562
wikipedia.org	41	437	0.6541	2	0.1691	526.22	#764978
zappos.com	105	4	0.4229	14	0.0088	362.24	#261016
google.com	6,068,897	737	0.6178	21,161	0.1475	344.87	#387579
blogger.com	1,354	31	0.4421	10	0.0245	261.51	#32562
youtube.com	2,082,754	194	0.6110	217	0.1438	221.45	#261016
wordpress.com	17,172,658	23,769	0.5427	439,531	0.0984	192.85	#261016
amazon.com	565,767	119	0.5326	29,730	0.1026	184.43	#764978
flickr.com	4,961,041	188	0.5071	775	0.0673	142.93	#456285
nytimes.com	2,566,437	390	0.5466	10,123	0.1128	141.03	#32562
yahoo.com	2,591,234	678	0.5297	28,841	0.0898	137.91	#235325
blogmura.com	254,341	278	0.3848	783	0.0027	128.84	#76398
myspace.com	2,310,596	96	0.4891	20,993	0.0623	100.69	#122579
msplinks.com	0	2	0.3976	0	0.0043	83.82	#122579
bit.ly	1,336	5	0.4395	14	0.0224	83.26	#261016
apple.com	311,408	76	0.4786	481	0.0502	82.66	#764978
facebook.com	666,461	54	0.5012	13	0.0724	80.79	#235325
microsoft.com	972,037	56	0.4702	118	0.0431	77.18	#764978
tinyurl.com	877,296	36	0.4647	761	0.0458	67.90	#261016
typepad.com	5,535,530	7,488	0.4955	142,460	0.0697	67.51	#518120
feedburner.com	333,844	49	0.4389	403	0.0199	67.05	#32562
igooi.com	0	247	0.3902	0	0.0034	63.20	#92825
adobe.com	1,484,614	68	0.4495	181	0.0277	61.96	#60228
topix.net	291,056	48	0.4291	2,791	0.0292	54.47	#504611
bbc.co.uk	1,123,960	107	0.5022	2,161	0.0749	52.74	#32562
imdb.com	689,754	39	0.4849	34	0.0592	51.33	#504611

DOMAIN	BETWEENNESS	BRIDGES	CLOSENESS	DEGREE	EIGENVECTOR	PAGERANK	COMMUNITY
oyax.com	0	136	0.4191	0	0.0078	49.72	#92825
washingtonpost.com	458,785	99	0.4906	3,172	0.0755	46.29	#32562
americablog.com	22,835	21	0.3966	5,367	0.0057	45.16	#32562
msn.com	459,185	139	0.4903	7,144	0.0704	41.32	#504611
guardian.co.uk	611,728	438	0.4908	19,697	0.0685	40.21	#396733
cnn.com	1,217,548	156	0.4992	48,487	0.0758	39.53	#504611
digg.com	155,536	31	0.4494	53,061	0.0261	38.98	#235325
wsj.com	1,308,752	62	0.4910	2,003	0.0736	38.15	#32562
cnet.com	404,717	190	0.4598	8,934	0.0390	36.73	#764978
go.com	171,523	73	0.4819	1,753	0.0665	35.64	#504611
creativecommons.org	148,583	24	0.4347	142	0.0168	34.17	#122579
fc2.com	769,927	1,004	0.4067	34,034	0.0062	33.68	#76398
vimeo.com	224,657	21	0.4495	5	0.0285	33.38	#32562
technorati.com	0	10	0.4363	0	0.0176	32.10	#32562
live.com	1,304,897	1,078	0.4465	19,854	0.0290	28.82	#504611
latimes.com	110,451	105	0.4784	1,932	0.0647	26.29	#32562
dailykos.com	8,965	14	0.4217	2,452	0.0195	25.74	#32562
cnbc.com	3,248	7	0.4247	157	0.0183	24.89	#32562
ameblo.jp	921,422	4,930	0.3939	235,651	0.0040	24.59	#76398
icio.us	0	0	0.4331	0	0.0153	23.29	#235325
reuters.com	414,374	46	0.4817	2,727	0.0624	22.89	#32562
cdc.gov	465	6	0.4519	38	0.0424	22.15	#32562
photobucket.com	220	5	0.4400	4	0.0202	22.08	#32562

How each value was computed every score written back as a node property via Cypher

Betweenness `gds.betweenness`

Brandes on the directed graph, `samplingSize:2000`.
Bridges of information flow — **blogspot.com** and **wordpress.com** dominate.

Bridges `gds.bridges`

Edges whose removal disconnects the (undirected) graph. Per-domain count of incident bridges — huge for platforms that fan out to isolated blogs.

Closeness `gds.closeness`

Wasserman–Faust variant on the degree ≥ 50 core (exact all-pairs is $O(V \cdot E)$). Reference hubs like **wikipedia.org** sit closest to everything.

Degree `gds.degree`

Weighted out-degree = Σ link weight. Raw outbound volume; **blogspot.com** alone emits 7.2M links.

Eigenvector `gds.eigenvector`

Influence weighted by the influence of who links you. Converged in 46 iterations.

PageRank `gds.pageRank`

Random-surfer importance, weighted, damping 0.85. **twitter.com** tops it — 2009 was its breakout.

Louvain `gds.louvain`

Modularity community detection on the undirected weighted graph — 3,145 communities at modularity 0.61.

Source: Stanford SNAP MemeTracker (quotes_2009-04) • engine: Neo4j 2026.05 + Graph Data Science 2026.05, Docker `neo4j-gds` • graph: 795,268 domains / 2,213,774 weighted links • betweenness sampled (2,000 pivots); closeness on the degree ≥ 50 core (10,290 nodes).